

Q1. Comparing Distributions – Histogram, Density and ECDF

```
library(ggplot2)
```

```
Time <- c(18, 21, 22, 24, 25, 25, 27, 28, 29, 30,  
          31, 32, 33, 35, 36, 38, 41, 43, 46, 50)
```

```
data <- data.frame(Time)
```

```
ggplot(data, aes(x = Time)) +
```

```
  geom_histogram(  
    bins = 7,  
    fill = "skyblue",  
    color = "black"
```

```
  ) +
```

```
  labs(  
    title = "Histogram of Processing Times",  
    x = "Processing Time (minutes)",  
    y = "Frequency"
```

```
  ) +
```

```
  theme_minimal()
```

```
ggplot(data, aes(x = Time)) +
```

```
  geom_density(  
    fill = "lightblue",  
    alpha = 0.5
```

```
  ) +
```

```
  labs(  
    title = "Density Plot of Processing Times",  
    x = "Processing Time (minutes)",  
    y = "Density"
```

```
  ) +
```

```
  theme_minimal()
```

```
ggplot(data, aes(x = Time)) +
```

```
  stat_ecdf(  
    fill = "lightblue",  
    alpha = 0.5
```

```
  ) +
```

```
  labs(  
    title = "ECDF of Processing Times",  
    x = "Processing Time (minutes)",  
    y = "Cumulative Frequency"
```

```
  ) +
```

```
  theme_minimal()
```

```
ggplot(data, aes(x = Time)) +
```

```
  stat_ecdf(  
    fill = "lightblue",  
    alpha = 0.5
```

```
  ) +
```

```
  labs(  
    title = "ECDF of Processing Times",  
    x = "Processing Time (minutes)",  
    y = "Cumulative Frequency"
```

```
  ) +
```

```
  theme_minimal()
```

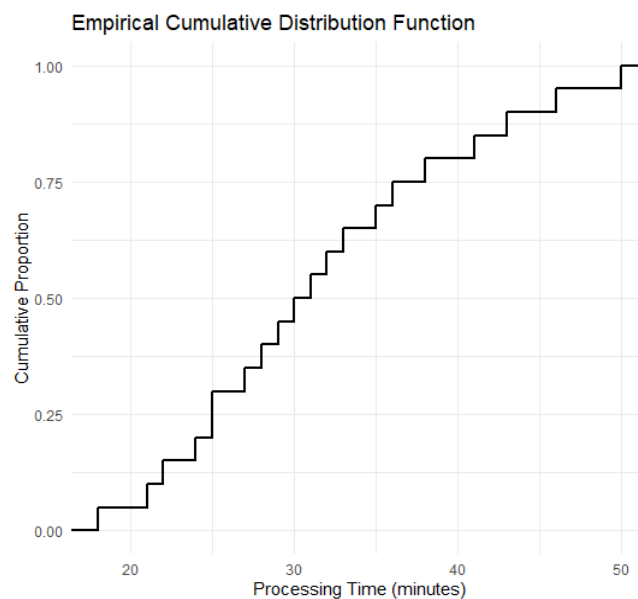
```
ggplot(data, aes(x = Time)) +
```

```
  stat_ecdf(  
    fill = "lightblue",  
    alpha = 0.5
```

```

geom = "step",
linewidth = 1
) +
labs(
  title = "Empirical Cumulative Distribution Function",
  x = "Processing Time (minutes)",
  y = "Cumulative Proportion"
) +
theme_minimal()

```



Q2. Q-Q Plot and Normality Analysis

```

Response <- c(42, 45, 47, 49, 50, 51, 52, 53, 54, 55,
              56, 57, 58, 60, 61, 63, 66, 70, 76, 85)

```

```

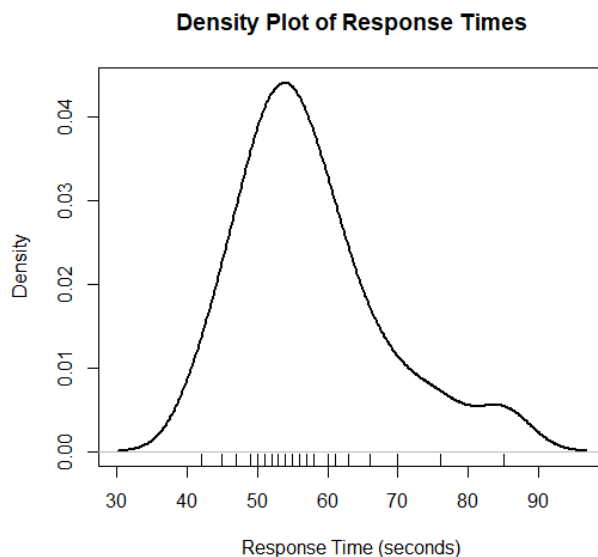
qqnorm(Response,
  main = "Normal Q-Q Plot of Response Times",
  xlab = "Theoretical Quantiles",
  ylab = "Observed Quantiles",
  pch = 19)
qqline(Response,
  col = "red",
  lwd = 2)

```

```

hist(Response,
      breaks = 7,
      main = "Histogram of Response Times",
      xlab = "Response Time (seconds)",
      ylab = "Frequency")
plot(density(Response),
     main = "Density Plot of Response Times",
     xlab = "Response Time (seconds)",
     ylab = "Density",
     lwd = 2)
rug(Response)

```



Q3. Visualizing Many Distributions at Once

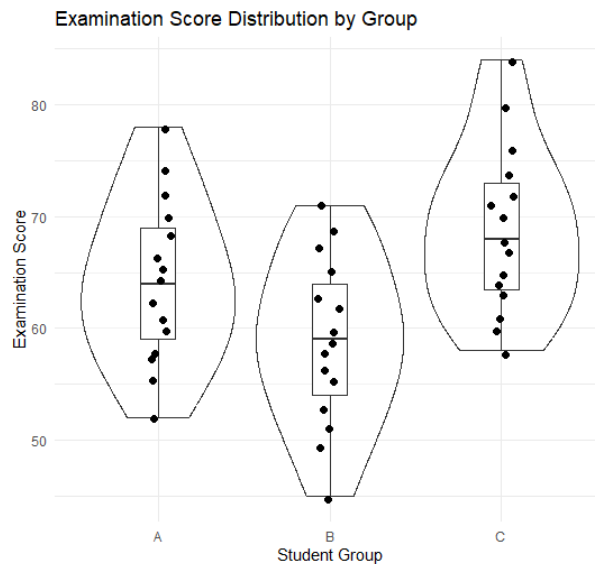
```

library(ggplot2)
Group_A <- c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78)
Group_B <- c(45,49,51,53,55,56,58,59,60,62,63,65,67,69,71)
Group_C <- c(58,60,61,63,64,65,67,68,70,71,72,74,76,80,84)
scores <- data.frame(
  Group = rep(c("A", "B", "C"), each = 15),
  Score = c(Group_A, Group_B, Group_C)
)

```

```
print(scores)
```

```
ggplot(scores, aes(x = Group, y = Score)) +  
  geom_violin(alpha = 0.4) +  
  geom_boxplot(width = 0.2) +  
  geom_jitter(width = 0.08, size = 2) +  
  labs(  
    title = "Examination Score Distribution by Group",  
    x = "Student Group",  
    y = "Examination Score"  
  ) + theme_minimal()
```



Q4. Relationships and Multivariate Visualization

```
library(ggplot2)
```

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38)
```

```
Visits <- c(120,135,150,165,180,195,210,225,245,260,  
           275,290,305,320,340)
```

```
Sales <- c(48,50,53,55,58,61,63,66,69,71,  
          74,77,80,83,86)
```

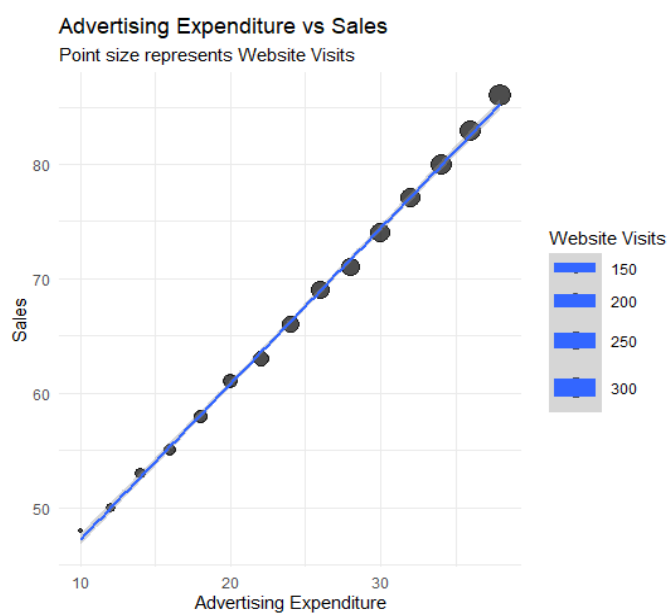
```
data <- data.frame(  
  Advertising,
```

```
  Advertising,
```

```

Visits,
Sales
)
ggplot(data, aes(x = Advertising,
                 y = Sales,
                 size = Visits)) +
  geom_point(alpha = 0.7) +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  labs(
    title = "Advertising Expenditure vs Sales",
    subtitle = "Point size represents Website Visits",
    x = "Advertising Expenditure",
    y = "Sales",
    size = "Website Visits"
  ) +
  theme_minimal()

```



Q5. Misleading Visualization and Redesign

```

library(ggplot2)

Department <- c("CSE", "ECE", "EEE", "MECH",
               "CIVIL", "AIDS", "IT")
Pass_Percentage <- c(91, 93, 92, 89, 90, 95, 94)
data <- data.frame(Department, Pass_Percentage)
ggplot(data, aes(x = Department, y = Pass_Percentage)) +
  geom_col(width = 0.9) +
  coord_cartesian(ylim = c(85, 96)) +
  labs(
    title = "Misleading Visualization of Pass Percentage",
    x = "Department",
    y = "Pass Percentage (%)"
  ) +
  theme_minimal()
data_ordered <- data[order(-data$Pass_Percentage), ]
ggplot(data_ordered,
       aes(x = reorder(Department, Pass_Percentage),
           y = Pass_Percentage)) +
  geom_col(width = 0.6) +
  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 10)
  ) +
  geom_text(
    aes(label = paste0(Pass_Percentage, "%")),
    vjust = -0.3
  ) +
  labs(
    title = "Pass Percentage by Department",
    x = "Department",

```

y = "Pass Percentage (%)"

) +

theme_minimal()

